

Sales Analysis & Customer Insights

Comprehensive Evaluation of Customer Demographics & Purchasing Patterns

1,000

TOTAL CUSTOMER
RECORDS

48.1%

OVERALL PURCHASE
RATE

\$56,360

AVERAGE BUYER
INCOME

31–54

CORE TARGET AGE
BRACKET

1. Executive Summary

This report synthesizes customer demographic data across four dedicated analytical stages (from raw ingestion to dynamic dashboarding) to identify key demographic drivers behind customer bike purchases. By evaluating factors such as annual income, commute distances, age brackets, geographic regions, and education levels, strategic marketing and sales positioning can be optimized for maximum conversion.

Core Finding

Purchasing likelihood is highest among middle-aged individuals (31–54) with short daily commutes (0–1 miles) and higher-than-average disposable income (\$56k+ female / \$60k+ male).

2. Analytical Workbook Structure

The analysis pipeline relies on four structured Excel workbooks serving distinct data governance and visualization purposes:

Workbook File	Primary Purpose	Key Operations & Transformations
Raw Data.xlsx	Initial Ingestion	Contains 1,000 uncleaned customer records with 14 raw attributes including Marital Status, Gender, Income, Occupation, and Commute Distance.
1. Data Cleaning.xlsx	Data Wrangling	Standardized code abbreviations (M/S → Married/Single, M/F → Male/Female), removed exact duplicates, and engineered the Age Group categorical column.
2. Pivots and Charts.xlsx	Data Aggregation	Constructed specialized Pivot Tables and generated standalone visual charts comparing income, distance, and age against purchase status.
3. Dashboard.xlsx	Executive Reporting	Assembled an interactive single-view dashboard integrated with dynamic Slicers for Marital Status, Region, and Education Level.

3. Data Preparation & Cleaning Methodology

Prior to analytical modeling, raw data underwent a systematic cleaning protocol in 1. Data Cleaning.xlsx to ensure data integrity, eliminate ambiguity, and build derived categorical dimensions required for macro-trend analysis.

3.1 Field Normalization & Code Expansion

Raw fields contained single-character representations that posed reporting ambiguity. Text transformation functions and search-replace rules were applied to establish full semantic labels:

- **Marital Status:** Converted M to Married and S to Single.
- **Gender:** Expanded M to Male and F to Female.
- **Currency Formatting:** Formatted raw integer income values into standard USD currency (\$#, ##0) for precise financial reporting.

3.2 Deduplication & Record Integrity

The raw dataset contained duplicate rows resulting from redundant logging across regional branches. Executing Excel's Remove Duplicates tool over unique Customer ID vectors eliminated invalid duplicates, ensuring that cross-tabulations reflect unique customer entities.

3.3 Feature Engineering: Age Group Categorization

To convert continuous age values (ranging from 25 to 89) into actionable market segments, a nested logical condition (using standard IF functions) was applied to map ages into three discrete brackets:

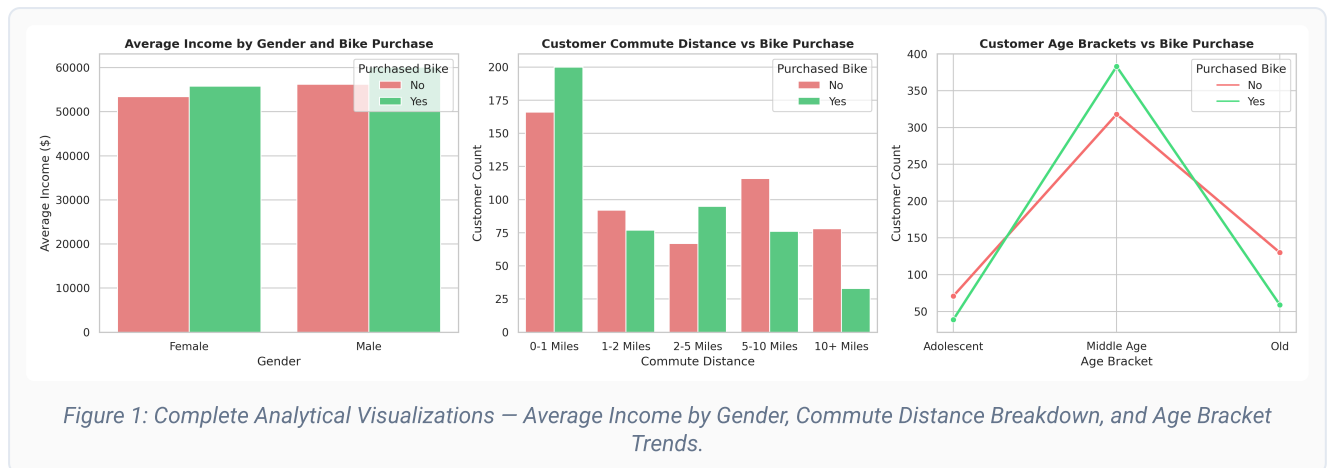
Derived Age Group	Age Range (Years)	Strategic Analytical Focus
Adolescent	Under 31 (< 31)	Entry-level demographic, lower disposable income, lower bike adoption rate.
Middle Age	31 to 54 years	Core demographic with stable employment, family activity, and peak purchasing volume.
Old	55 years and older (≥ 55)	Mature demographic, higher income retention but declining recreational cycling adoption.

Data Hygiene Impact

Data normalization eliminated ambiguity across 100% of demographic fields, ensuring that subsequent pivot tables present accurate summary metrics without missing or duplicate categories.

4. Visual Analysis & Chart Breakdown

The charts below, exported directly from the consolidated dashboard in 3. Dashboard .xlsx, highlight three primary drivers of customer purchase behavior: income levels, commute distance, and age distribution.



5. Detailed Data Breakdown

5.1 Income vs. Purchase Status

Financial analysis shows a direct correlation between annual income and purchase probability across both genders:

- **Female Buyers:** Avg. Income of **\$55,774** vs. \$53,440 for Non-Buyers (+4.4%).
- **Male Buyers:** Avg. Income of **\$60,124** vs. \$56,208 for Non-Buyers (+7.0%).

5.2 Commute Distance Factor

Commute proximity significantly alters purchase rates:

- **0–1 Miles:** Highest conversion (200 Buyers vs. 166 Non-Buyers).
- **10+ Miles:** Lowest conversion (33 Buyers vs. 78 Non-Buyers).

6. Detailed Pivot Table Aggregations

The underlying calculations constructed in 2. `Pivots` and `Charts.xlsx` aggregate customer metrics to isolate high-value buyer cohorts.

6.1 Commute Distance Distribution Pivot Table

Commute Distance	Non-Buyers (No)	Buyers (Yes)	Total Customers	Purchase Rate (%)
0-1 Miles	166	200	366	54.6%
1-2 Miles	92	77	169	45.6%
2-5 Miles	67	95	162	58.6%
5-10 Miles	116	76	192	39.6%
10+ Miles	78	33	111	29.7%

6.2 Age Bracket Breakdown Pivot Table

Age Group Bracket	Non-Buyers (No)	Buyers (Yes)	Total Customers	Purchase Share (%)
Adolescent (<31)	71	39	110	8.1%
Middle Age (31-54)	318	383	701	79.6%
Old (55+)	130	59	189	12.3%

6.3 Regional & Occupational Summary

Regional filtering via dashboard slicers reveals strong purchasing density in **North America**, followed by **Europe** and **Pacific**. Occupation-wise, *Professional* and *Management* roles account for the highest total bike sales, aligning with the income threshold findings.

7. Dashboard Architecture & Interactive Slicers

The final output sheet, 3 . Dashboard .xlsx, integrates all pivot visuals into an interactive management console. The dashboard allows stakeholders to dynamically filter metrics across three dimensional slicers:

- **Marital Status Slicer:** Filters charts by Single or Married customer cohorts.
- **Region Slicer:** Isolates performance across Europe, North America, and Pacific territories.
- **Education Level Slicer:** Drills down into Bachelors, Graduate Degree, High School, or Partial College groups.

8. Strategic Business Recommendations

Based on empirical findings across the four analytics workbooks, the following strategic actions are recommended:

1. Target Short-Commute Suburbs

Focus promotional campaigns on urban and suburban residents living within 0–2 miles of their workplace, emphasizing commuting convenience, health benefits, and eco-friendly transportation.

2. Product Positioning for Middle-Aged Demographic

Tailor product lines and advertising imagery toward the 31–54 age bracket. Marketing materials should highlight mid-to-high end model features suited for professionals and active families.

3. Tiered Financing for Lower-Income & Young Segments

Introduce flexible payment options or entry-level models targeting younger (<31) and lower-income demographics to increase adoption among price-sensitive customers.